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Bihar Engineering University, Patna

B.Tech 2nd Semester Exam., 2022

Time: 3 hours

Code: 100203 ***Subject: Chemistry*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. I is compulsory.

Q.1 Answer in brief any seven of the following: **[2x7=14]**

(a) State de Broglie's principle.

(b) What is an orbital?

(c) Arrange the covalent bonds C—F, C—Br, C—Cl and C—I in increasing order of strength.

(d) Which of NH_3, PH_3, AsH_3 and SbH_3 possesses largest bond angle? Give reason.

(e) Out of Cr^{2+} and Cr^{3+}, which one is stable in aqueous solution?

(f) Cu^{+} is colourless, but Cu^{2+} is coloured. Why?

(g) What is selection rule?

(h) A gas expands against vacuum. What is the work done on it?

(i) What is the condition for a reaction to be in equilibrium?

(j) The presence of CO_2 in boiler feed water should be avoided. Why?

Q.2 Solve both questions : **[5+5+4=14]**

(a) Write short notes on the following:
(i) Wave-particle duality (ii) Heisenberg's uncertainty principle

(b) Discuss photoelectric effect (with mathematical equations) for explaining the particle nature of light.

(c) How many photons of light having a wavelength of 4000 Å are necessary to provide 1 J of energy? ($h = 6.626\times10^{-34}$ J s)

Q.3 Solve this question : **[6+8=14]**

(a) Which of the following two molecules has a higher bond length? (i) O_2 (ii) O_2^{+} (iii) O_2^{-} Explain using molecular orbital theory.

(b) Draw the MO energy level diagram for NO molecule. Using this diagram, calculate and explain bond order and magnetic behaviour of (i) NO, (ii) NO^{+} and (iii) NO^{-}.

Q.4 Solve all parts : **[5+5+4=14]**

(a) The internuclear distance of NaCl is 2.36×10^{-10}m. Calculate the reduced mass and moment of inertia of NaCl. (Atomic mass of Cl = 35×10^{-3}kg mol^{-1} and Na = 23×10^{-3}kg mol^{-1}).

(b) Calculate the force constant for CO, if it absorbs at 2.143×10^{5}m^{-1}. (Atomic mass of C = 12×10^{-3}kg mol^{-1} and O = 16×10^{-3}kg mol^{-1})

(c) Which type of electronic transition(s) is/are observed in UV-visible spectrum of phenol in the range 200 nm to 900 nm? Justify your answer.

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Q.5 Solve all parts : **[5+4+5=14]**

(a) How many types of ^1H NMR signals are expected for (i) CH_2BrCH_2Cl and (ii) CH_3OCH_3? Mention relative intensity ratio for the signal(s) observed for (i) and (ii).

(b) At what frequency shift from TMS, would a group of nuclei with δ = 1:5 resonate in NMR spectrometers operating at 60 MHz and 400 MHz?

(c) 2 mol of NH_3 at 300 K occupy a volume of $5\times10^{-3}m^3$. Calculate the pressure using van der Waals equation ($a = 0.417 Nm^4mol^{-2}$ and $b = 0.037\times10^{-3}m^3mol^{-1}$). Compare the above result with the pressure calculated using ideal gas equation.

Q.6 Solve both questions : **[7+7=14]**

(a) 7 mol of a monatomic ideal gas are compressed reversibly and adiabatically. The initial volume is 16 dm^3 and the final volume is 7 dm^3. The initial temperature is 27 °C.

(i) What would be the final temperature in this process?

(ii) Calculate w, q and ΔU for the process.

Given, $C_v = 20{\cdot}91 JK^{-1}mol^{-1}$, $\gamma = 1{\cdot}4$.

(b) Write a cell (in proper cell representation) whose cell reaction is $AgCl \rightarrow Ag^+ + Cl^-$, using the following standard electrode potentials at 298 K:

$E°_{AgCl/AgCl^-} = 0{\cdot}22V$, $E°_{Ag^+/Ag} = 0{\cdot}80V$

Calculate $E°$ of the cell. Calculate solubility product (or solubility constant) of AgCl and its solubility at 298 K.

Q.7 Solve both questions : **[10+4=14]**

(a) A water sample had the following constituents per litre:

$CaCO_3 = 65mg$, $MgHCO_3 = 80mg$

$CaSO_4 = 155mg$, $MgSO_4 = 135mg$

$NaCl = 8mg$

Calculate the quantity of temporary and permanent hardness in the water sample. Calculate the quantity of lime (80% purity) and soda (90% purity) required for softening of 1 million litre of above water sample.

(b) In an experiment to determine the hardness of a sample of water, 25 ml of N/50 Na_2CO_3 solution was added to 100 ml of water sample. After complete precipitation of insoluble carbonate, the unreacted Na_2CO_3 was titrated against N/50 H_2SO_4 solution, when 10 ml of acid was required. Calculate the hardness and comment on the nature of hardness so determined.

Q.8 Solve both questions : **[8+6=14]**

(a) Write notes on the following:

(i) Optical isomerism of lactic acid

(ii) Optical isomerism of tartaric acid

(b) Differentiate between the following:

(i) Enantiomers and diastereomers

(ii) Racemic mixture and meso-compounds

Q.9 Solve both questions : **[8+6=14]**

(a) Explain the following:

(i) Acetic acid is stronger acid than phenol.

(ii) The amino group in aniline is o- and p-directing but amide group is meta-directing.

(b) Arrange the following carbocations in order of increasing stability with suitable reasons:

$C_6H_5CH_2^+$

$(CH_3)_3C^+$

$(C_6H_5)_2CH^+$

$CH_3CH_2^+$

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